

## Tangents to polar curves — 7.4

If we have a function of the form  $r = F(\theta)$ , we get a parametrization by converting to Cartesian coordinates.

Any point on the curve is of the polar form  $(f(\theta), \theta)$ , and in Cartesian coordinates, this is  $(r \cos \theta, r \sin \theta)$

$$= (f(\theta) \cos(\theta), f(\theta) \sin \theta)$$

Example: Find the line tangent to  $r = 2\cos\theta$  at the (polar) point  $(\sqrt{3}, \frac{\pi}{6})$ .

→ parameterize the curve as  $(2\cos\theta \cdot \cos\theta, 2\cos\theta \cdot \sin\theta)$

$$x(\theta) = 2\cos^2\theta \quad \Rightarrow \quad x'(\theta) = 4\cos\theta \cdot -\sin\theta$$

$$y(\theta) = 2\cos\theta \sin\theta = \sin(2\theta) \quad y'(\theta) = -2\sin\theta \sin\theta + 2\cos\theta \cos\theta = 2(\cos^2\theta - \sin^2\theta) = 2\cos(2\theta)$$

$$x'(\frac{\pi}{6}) = 4 \cdot \frac{\sqrt{3}}{2} \cdot -\frac{1}{2} = -\sqrt{3}$$

slope is then  $-\frac{1}{\sqrt{3}}$

$$y'(\frac{\pi}{6}) = 2 \cdot \cos(\frac{\pi}{3}) = 2 \cdot \frac{1}{2} = 1$$

Line passes through  $(2\cos^2(\frac{\pi}{6}), 2\cos(\frac{\pi}{6})\sin(\frac{\pi}{6})) = (\frac{3}{2}, \frac{\sqrt{3}}{2})$

$$\text{so is } y - \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x - \frac{3}{2}) \Leftrightarrow y = \sqrt{3} - \frac{x}{\sqrt{3}}.$$